Physical Chemistry CUME
April 4, 2014

At Earth's distance from the Sun, the intensity of sunlight is about 1350 W/m^2 outside the atmosphere. The dominant wavelength is about 500 nm. To simplify the calculations, assume in what follows that *all* photons emitted by the Sun have a wavelength of 500 nm. Photons, just like heavy particles, have mechanical momentum, given by h/λ where h is Planck's constant (h=6.62608E-34 J/s) and λ is the wavelength.

(a) What is the photon flux in $m^{-2}s^{-1}$, i.e. the number of photons which hit a 1 m^2 area perpendicular to sunlight at the Earth's distance from the Sun per second? (30 p)

(b) What is the force (in N) arising from light pressure on a 1 m^2 absorbing surface of a spacecraft, perpendicular to the Sun's rays, at the Earth's distance from the Sun? (20 p)

(c) According to the theory of thermal radiation, the most intense electromagnetic wave radiated by a black body at absolute temperature T has a wavelength $\lambda_{max} = hc/(5kT)$. Here k is the Boltzmann constant, k=1.38066E-23 J/K , and c is the speed of light, 2.9979E8 m/s. The Sunlight (outside the atmosphere) has its maximum intensity λ_{max} = 470 nm. Assuming that the Sun is a perfect (black-body) radiator, which is largely true, estimate the temperature of the Sun's surface. (Note: on the surface, the most intense color in the Sun's spectrum is at slightly higher wavelength, about 500 nm, because the blue light is scattered and absorbed more than the longer wavelengths). (20 p)

(d) The (average) distance of Earth from the Sun is about 150 million km. How much energy does the Sun radiate per second (what is its power)? (20 p)

(e) What is the color of light with a wavelength of 500 nm? (10 p)

Help:

(a) Calculate he energy of a single photon, and from the total energy per second calculate tye number of photons.
(b) The force is the change in the momentum per unit time [F=ma=d(mv)/dt]. Calculate the total momentum of the photons hitting the surface in 1 s, and divide the result by 1 s to get the right dimension.
(d) Calculate the surface area of a large sphere which has a radius of the Sun-Earth distance. All light the Sun emits goes through this, and each square meter receives 1350 W. The formula for the surface of the sphere is $4\pi r^2$ where r is its radius, a formula first derived by Archimedes.